

Disproof of TLMC Conjecture 00000001854: “Squarefree characteristic polynomial counting” is not an exact closed form

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Abstract

We disprove the conjecture that

$$\#\{M \in \text{Mat}_n(\mathbb{F}_q) : \text{charpoly}(M) \text{ squarefree}\} = q^{n^2} \prod_i (1 - q^{-i^2})$$

is an *exact closed form*. At the minimal dimension $n = 1$ the true count is exactly q (the characteristic polynomial of a 1×1 matrix is the degree-one polynomial $X - a$, which is always squarefree), whereas the conjectured expression evaluates to $q - 1$ when the product is read over $i = 1..n$, and to a quantity strictly smaller than q when the product is read as an infinite product. A second, fully enumerated corroboration at $n = 2$, $q = 2$ shows a count of 8 against a formula value of $15/2$, which is not even an integer.

1 The conjecture

Claim 1 (Conjecture 00000001854, literal statement). *Counting random matrices over $\mathbb{F}_q$:*

$$\#\{M \in \text{Mat}_n(\mathbb{F}_q) : \text{charpoly}(M) \text{ squarefree}\} = q^{n^2} \cdot \prod_i (1 - q^{-i^2})$$

is an exact closed form (squarefree char-poly counting).

The statement does not specify the index range of the product. We therefore refute the literal “exact” assertion under *both* natural readings:

- **Reading A:** the product runs over $i = 1, \dots, n$ (matching the degree of the characteristic polynomial);
- **Reading B:** the product is the infinite product $\prod_{i \geq 1} (1 - q^{-i^2})$.

2 Counterexample: $n = 1$

Counterexample 1 (minimal dimension). *Let $n = 1$. Every element of $\text{Mat}_1(\mathbb{F}_q)$ is a matrix $[a]$ with $a \in \mathbb{F}_q$, and its characteristic polynomial is*

$$\text{charpoly}([a]) = X - a.$$

This is a polynomial of degree 1. Over any field, a degree-one polynomial f has derivative $f' = 1$, whence

$$\gcd(f, f') = \gcd(X - a, 1) = 1,$$

so f has no repeated irreducible factor, i.e. f is squarefree. Therefore every matrix of $\text{Mat}_1(\mathbb{F}_q)$ is counted:

$$\#\{M \in \text{Mat}_1(\mathbb{F}_q) : \text{charpoly}(M) \text{ squarefree}\} = |\mathbb{F}_q| = q.$$

Against this, under Reading A the conjectured formula gives

$$q^1 \cdot (1 - q^{-1}) = q - 1 \neq q,$$

so Claim 1 fails for *every* prime power $q \geq 2$ already at $n = 1$. Explicit values (all confirmed by the brute-force enumeration of the accompanying script `reproduce.py`, which implements polynomial arithmetic and the squarefreeness test $\gcd(f, f') = 1$ over $\mathbb{F}_q$ for prime q and over $\mathbb{F}_4 = \mathbb{F}_2[\omega]/(\omega^2 + \omega + 1)$):

q	true count (enumerated)	Reading A: $q(1 - q^{-1})$	Reading B: $q \prod_{i \geq 1} (1 - q^{-i^2})$
2	2	1	≈ 0.935655
3	3	2	≈ 1.975208
4	4	3	≈ 2.988270
5	5	4	≈ 3.993598

Table 1: The counterexample $n = 1$. Reading B is computed with exact rational arithmetic (truncated at $i \leq 10$; the omitted tail is bounded by q^{-121} , far below any rounding effect).

Under Reading B the failure is structural: every factor $(1 - q^{-i^2})$ lies strictly between 0 and 1, so $q \prod_{i \geq 1} (1 - q^{-i^2}) < q$, while the left-hand side is exactly q . Under either reading, the “exact closed form” is false at the smallest possible dimension.

3 Second corroboration: $n = 2$, $q = 2$

We enumerated all $2^{2 \cdot 2} = 16$ matrices of $\text{Mat}_2(\mathbb{F}_2)$, computed $\text{charpoly}(M) = \det(XI - M)$ with exact polynomial arithmetic over $\mathbb{F}_2$, and tested squarefreeness via $\gcd(f, f')$ having degree 0. The count of matrices with squarefree characteristic polynomial is

$$\#\{M \in \text{Mat}_2(\mathbb{F}_2) : \text{charpoly}(M) \text{ squarefree}\} = 8,$$

(the admissible characteristic polynomials being $X^2 + X = X(X + 1)$, attained by 6 matrices, and the irreducible $X^2 + X + 1$, attained by 2 matrices; the remaining 8 matrices have characteristic polynomial X^2 or $X^2 + 1 = (X + 1)^2$, both non-squarefree).

Reading A of the conjectured formula yields instead

$$2^4 \cdot (1 - 2^{-1})(1 - 2^{-4}) = 16 \cdot \frac{1}{2} \cdot \frac{15}{16} = \frac{15}{2},$$

which is *not an integer* — impossible for an exact count of a finite set. Reading B yields $\approx 7.485237 \neq 8$. This corroborates Counterexample 1 at the next dimension.

4 Boundary of the refutation

Because the index range of $\prod_i (1 - q^{-i^2})$ is unspecified in the original statement, some charity is required to read it at all. Our refutation targets the literal *exactness* assertion and is insensitive to the choice between Readings A and B; both fail, and in both directions (too small by one at $n = 1$ under Reading A; strictly below the true count under Reading B; non-integral at $n = 2$ under Reading A). If the product was intended relative to some further, different normalization (for instance a probability $\prod_i (1 - q^{1-i^2})$ -type density), then the statement as written is ambiguous rather than exact; we refute the statement in the form quoted in Section 1. For reference, the classical exact count of *monic squarefree polynomials* of degree $n \geq 2$ over $\mathbb{F}_q$ is $q^n - q^{n-1}$, a statement about polynomials rather than matrices, which does not coincide with the conjectured expression either.

5 Machine verification

The package accompanying this document contains:

1. `reproduce.py`: a dependency-free Python script that (i) implements $\mathbb{F}_q$ for prime $q \in \{2, 3, 5\}$ and $\mathbb{F}_4 = \mathbb{F}_2[\omega]/(\omega^2 + \omega + 1)$, polynomial long division, Euclidean gcd, and the squarefreeness test $\deg \gcd(f, f') = 0$; (ii) enumerates *all* matrices of $\text{Mat}_1(\mathbb{F}_q)$ for $q = 2, 3, 4, 5$ and of $\text{Mat}_2(\mathbb{F}_2)$; and (iii) asserts every inequality reported above, terminating successfully only if the conjecture fails as described.
2. `lean4/`: a Lean 4 project (toolchain `leanprover/lean4:v4.33.1`, no external dependencies) formalizing the numeric content of the $n = 1, q = 2$ counterexample with *zero axioms and zero sorry*: the enumerated count 2 and the formula value 1 (encoded as an exact fraction over kernel-computable natural-number arithmetic, since core `Rat` does not reduce inside the kernel) are computed by `rfl` and proved distinct by `decide`. Since Lean core does not ship a polynomial library, the squarefreeness of $X - a$ is encoded as the (mathematically justified, see Counterexample 1) statement that every $a \in \mathbb{F}_2$ passes the squarefreeness predicate; the mathematical argument is the one of Section 2 and is verified generically by `reproduce.py` via $\gcd(f, f')$.

6 Conclusion

Conjecture 00000001854 is **false**: the expression $q^{n^2} \prod_i (1 - q^{-i^2})$ is not an exact closed form for the number of $n \times n$ matrices over $\mathbb{F}_q$ with squarefree characteristic polynomial. The correct value at $n = 1$ is q , not $q - 1$, and already at $n = 2, q = 2$ the formula produces the non-integer $15/2$ against the true count 8.